

Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE- Academic Year 2025-26

Compiler Construction Lab Batch 2022-26

Lab Assignment No: - 7	
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Batch	2022-26
Class	CSE B3
Academic Year & Semester	4 th Year, 7 th Semester
Date of Performance	
Title of Assignment	Postfix Expression Evaluation.
Practice Questions	1. YACC program for Postfix Expression Evaluation. 2. YACC program for Conversion of Infix to Postfix expression.

Source Code	1. pos.l <pre data-bbox="527 268 1401 716">%{ #include "pos.tab.h" #include <stdlib.h> }% %% [0-9]+ { yylval = atoi(yytext); return ID; } [+\\-*/] { return yytext[0]; } [\\t\\r]+ ; \\n { return 0; } . { /* ignore other chars or handle errors */ } %%</pre>
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2. pos.y

```
%{
#include <stdio.h>
#include <assert.h>

void push(int val);
int pop(void);
int top(void);
int yylex(void);
void yyerror(const char *s);

%}

%token ID

%%

S : E { printf("= %d\n", top()); }
;

E : E E '+' { push(pop() + pop()); }
  | E E '-' { int temp = pop(); push(pop() - temp); }
  | E E '*' { push(pop() * pop()); }
  | E E '/' { int temp = pop(); push(pop() / temp); }
  | ID { push(yylval); }
;

%%

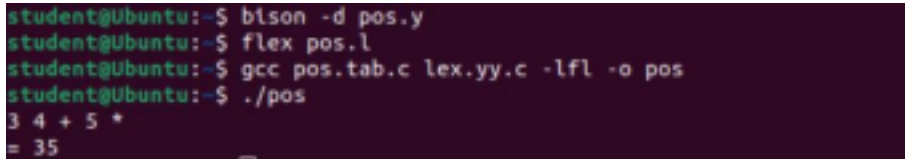
int st[100];
int i = 0;

void push(int val)
{
    assert(i < 100);
    st[i++] = val;
}

int pop(void)
{
    assert(i > 0);
    return st[--i];
}

int top(void)
{
    assert(i > 0);
    return st[i - 1];
}
```

	}
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	<pre>void yyerror(const char *s) { fprintf(stderr, "Error: %s\n", s); } int main(void) { return yyparse(); }</pre>
Output Screenshot	 <pre>student@ubuntu:~\$ bison -d pos.y student@ubuntu:~\$ flex pos.l student@ubuntu:~\$ gcc pos.tab.c lex.yy.c -lfl -o pos student@ubuntu:~\$./pos 3 4 + 5 * = 35</pre>
Post lab questions	1.YACC program for evaluating postfix expressions containing decimal numbers.
Conclusion	Thus we have implemented a YACC program for evaluation of postfix expressions and conversion from infix to postfix.